

D682

Artificial Intelligence Optimization for Computer Scientists

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

A team trains a gradient-boosted model on 50,000 support tickets using eight GPUs. Training accuracy is 99% and validation accuracy is 71%. The product owner wants it shipped by Friday. What is the most likely problem and fix?

- A. Nothing; validation accuracy is always lower, so ship it.
- B. Underfitting; add more trees and depth.
- C. Train longer on more GPUs.
- D. Overfitting; reduce model complexity, regularize, or use early stopping.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A model scores 62% on training data and 60% on validation data, both well below target. Which change is most likely to help?

- A. Add stronger regularization.
- B. Build a larger test set.
- C. Stop training earlier.
- D. Increase model capacity or add more informative features.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A warehouse wants a robot to learn the best picking route by trying routes and receiving a reward for faster picks. The warehouse stocks 120,000 SKUs and runs around the clock. Which AI approach fits?

- A. Reinforcement learning.
- B. A hand-coded rule for each of the 120,000 SKUs.
- C. Unsupervised clustering.
- D. Supervised classification.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A data scientist tries 200 combinations of learning rate and tree depth, keeps whichever scores best on the held-out test set, and reports that score as expected production accuracy. The data came from a well-regarded partner hospital. What is wrong?

- A. The partner hospital's data should be re-verified.
- B. 200 combinations is too few to find a good model.
- C. The test set was used for tuning, so the reported score is optimistic; tune on validation data and use the test set once.
- D. Nothing; the test set is the most honest measure.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

During training, the loss jumps around and gets larger each epoch instead of falling. The team uses a batch size of 64 on a new cloud cluster. What is the first adjustment to try?

- A. Train for more epochs.
- B. Move to a different cloud cluster.
- C. Raise the learning rate.
- D. Lower the learning rate.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A fraud model flags 200 of 1,000,000 transactions; 150 of the flagged ones are truly fraud. There were 300 fraudulent transactions in total. The model runs as a nightly batch job. What is the model's recall?

- A. 75%
- B. About 67%
- C. 50%
- D. About 99.98%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

A model's precision is 80% and it makes 400 positive predictions. How many of those are false positives?

- A. It cannot be known without recall.
- B. 80
- C. 320
- D. 20

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A model to detect a rare machine failure, which occurs in 2% of cases, reports 98% accuracy. It was built with a popular open-source library. What is the main concern?

- A. 98% is excellent; deploy it.
- B. A model that never predicts failure would also score 98%; check recall and precision for failures.
- C. The open-source library is unreliable.
- D. Accuracy must reach 100% for safety systems.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

Of 10,000 loans, 500 defaulted. A model predicts "no default" for every loan. What are its accuracy and its recall for defaults?

- A. 95% accuracy, 95% recall.
- B. 5% accuracy, 100% recall.
- C. 95% accuracy, 0% recall.
- D. 95% accuracy; recall cannot be computed.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A delivery app uses A* search to route drivers across a city of 3,000 intersections. An engineer proposes a heuristic that sometimes overestimates the remaining distance, to make searches faster. The app has a four-star rating. What is the effect?

- A. Searches may be faster, but A* is no longer guaranteed to find the shortest route.
- B. The rating will fall, so reject it.
- C. Searches are faster and still always optimal.
- D. A* is always slower with any heuristic.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A utility forecasts daily electricity demand from five years of data. An analyst randomly shuffles all days into an 80% training and 20% test split and gets very low test error. The data sits in a new lakehouse. What is the problem?

- A. Shuffling lets the model train on days after the test days; use a time-based split.
- B. The lakehouse may have corrupted the dates.
- C. Nothing; random splits always give unbiased estimates.
- D. The split should be 70/30.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

A model will predict next month's sales. The data is monthly from 2019 through 2025. Which split is appropriate?

- A. Randomly assign 80% of months to training.
- B. Train on 2019-2024; test on 2025.
- C. Train on 2021-2025; test on 2019-2020.
- D. Train and test on all months.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

A team trains a gradient-boosted model on 50,000 support tickets ~~using eight GPUs~~. Training accuracy is 99% and validation accuracy is 71%. ~~The product owner wants it shipped by Friday~~. What is the most likely problem and fix?

Target word: diagnose and fix **Decides:** A 28-point gap between training and validation.

A	Absolute language	A small gap is normal; 28 points is not.
B	Inverted operation	More complexity worsens overfitting.
C	Camouflage	Hardware does not decide the generalization gap; longer training tends to widen it.
D	Correct	The model memorized training data and does not generalize.

2. Answer D Concept 1 · transfer

A model scores 62% on training data and 60% on validation data, both well below target. Which change is most likely to help?

Target word: fix — reversed to underfitting **Decides:** Both scores are low and close together.

A	Inverted operation	Regularization limits capacity further.
B	Adjacent category	Improves the estimate but not the model.
C	Confusion set	Early stopping fights overfitting, not underfitting.
D	Correct	Underfitting needs a model able to capture more of the pattern.

3. Answer A Concept 2

A warehouse wants a robot to learn the best picking route by trying routes and receiving a reward for faster picks. ~~The warehouse stocks 120,000 SKUs and runs around the clock~~. Which AI approach fits?

Target word: approach selection **Decides:** Learning from trial and reward.

A	Correct	An agent learns actions that maximize reward through trial.
B	Camouflage	Reacts to the SKU count; hand-coding does not learn from rewards.
C	Confusion set	Groups similar items; it does not learn actions.
D	Adjacent category	Needs labeled correct answers, which the stem does not have.

4. Answer C Concept 3

A data scientist tries 200 combinations of learning rate and tree depth, keeps whichever scores best on the held-out test set, and reports that score as expected production accuracy. ~~The data came from a well-regarded partner hospital~~. What is wrong?

Target word: evaluation flaw **Decides:** The test set was used to choose the model.

A	Camouflage	Data source quality is not what the stem describes.
B	Adjacent category	Search size is not the flaw; more searching on the test set would worsen the leak.
C	Correct	Selection on the test set leaks it into the model choice.
D	Inverted operation	It stops being honest once it is used for selection.

5. Answer D Concept 4

During training, the loss jumps around and gets larger each epoch instead of falling. ~~The team uses a batch size of 64 on a new cloud cluster.~~ What is the first adjustment to try?

Target word: fix divergence **Decides:** Rising, erratic loss signals steps that are too large.

A	Dropped element	More epochs of a diverging run do not help.
B	Camouflage	Hardware does not cause the loss to rise.
C	Inverted operation	Larger steps diverge faster.
D	Correct	Smaller steps stop overshooting the minimum.

6. Answer C Concept 5

A fraud model flags 200 of 1,000,000 transactions; 150 of the flagged ones are truly fraud. There were 300 fraudulent transactions in total. ~~The model runs as a nightly batch job.~~ What is the model's recall?

Target word: recall **Decides:** TP = 150; actual positives = 300.

A	Confusion set	150 / 200 is precision.
B	Wrong stem number	200 / 300 compares flags to fraud cases, not true positives.
C	Correct	150 / 300 = 0.50.
D	Adjacent category	That is accuracy: (150 + 999,650) / 1,000,000.

7. Answer B Concept 5 · transfer

A model's precision is 80% and it makes 400 positive predictions. How many of those are false positives?

Target word: solve for false positives **Decides:** Precision = TP / all predicted positives.

A	Dropped element	Recall is not needed; precision and the prediction count suffice.
B	Correct	TP = 0.8 × 400 = 320; FP = 400 – 320 = 80.
C	Inverted operation	That is the true positives.
D	Wrong stem number	100 – 80 is a percent, not a count.

8. Answer B Concept 6

A model to detect a rare machine failure, which occurs in 2% of cases, reports 98% accuracy. ~~It was built with a popular open-source library.~~ What is the main concern?

Target word: evaluate accuracy on rare events **Decides:** 98% equals always predicting “no failure.”

A	Dropped element	Drops the class imbalance.
B	Correct	Accuracy hides whether any failures are caught.
C	Camouflage	The library does not decide the metric problem.
D	Absolute language	The problem is the metric, not its size.

9. Answer C Concept 6 · transfer

Of 10,000 loans, 500 defaulted. A model predicts “no default” for every loan. What are its accuracy and its recall for defaults?

Target word: compute both metrics **Decides:** It is right on 9,500 and catches 0 of 500 defaults.

A	Confusion set	Treats recall as accuracy.
B	Inverted operation	Describes a model that predicts “default” for every loan.
C	Correct	9,500 / 10,000 correct; 0 / 500 defaults found.
D	Dropped element	Recall is $0 / 500 = 0\%$.

10. Answer A Concept 7

A delivery app uses A* search to route drivers across a city of 3,000 intersections. An engineer proposes a heuristic that sometimes overestimates the remaining distance, to make searches faster. ~~The app has a four-star rating.~~ What is the effect?

Target word: effect on optimality **Decides:** The heuristic can overestimate.

A	Correct	Optimality requires an admissible heuristic.
B	Camouflage	The rating does not decide the algorithm property.
C	Dropped element	Drops the admissibility condition.
D	Absolute language	Good heuristics usually speed A* up.

11. Answer A Concept 8

A utility forecasts daily electricity demand from five years of data. An analyst randomly shuffles all days into an 80% training and 20% test split and gets very low test error. ~~The data sits in a new lakehouse.~~ What is the problem?

Target word: split flaw **Decides:** Random shuffling on time-ordered data.

A	Correct	Future information leaks into training and flatters the score.
B	Camouflage	Storage does not decide the leak.
C	Absolute language	Not for time series with trends and seasonality.
D	Wrong stem number	The ratio is not the issue; the ordering is.

12. Answer B Concept 8 · transfer

A model will predict next month’s sales. The data is monthly from 2019 through 2025. Which split is appropriate?

Target word: choose a split **Decides:** Test must come after training in time.

A	Confusion set	Leaks future months into training.
B	Correct	Mimics forecasting the future from the past.
C	Inverted operation	Tests on the past using the future.
D	Dropped element	No held-out data at all.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	6	
Wrong stem number	A figure copied from the prompt that is not the answer.	3	
Adjacent category	The right idea in the wrong bucket.	4	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	7	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	5	
Camouflage	Answers a stem fact that was never the question.	7	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Overfitting vs. underfitting	High training score with much lower validation score = overfitting (reduce complexity, regularize, add data); low scores on both = underfitting (add capacity or features).	Fixing overfitting by adding complexity, or underfitting by adding regularization.	A training score and a validation score side by side.
2	Matching the AI approach to the problem	Supervised learning needs labeled answers; unsupervised finds structure without labels; reinforcement learning learns actions from rewards.	Choosing supervised learning when there are no labels, only rewards.	"Learns by trying... receives a reward."
3	Keep the test set untouched	Tune on a validation set or cross-validation; use the test set once for the final estimate.	Picking hyperparameters by test-set score and reporting that score.	Many configurations compared "on the test set."
4	Learning rate	A learning rate that is too high makes loss bounce or grow; too low makes training slow.	Raising the rate or training longer when loss diverges.	Loss that jumps around or rises epoch to epoch.
5	Precision vs. recall	Precision = $TP / (TP + FP)$; recall = $TP / (TP + FN)$.	Swapping them, or quoting accuracy instead.	Counts of flagged cases and actual positives.
6	Accuracy misleads on imbalanced data	When one class is rare, a model that always predicts the common class scores high accuracy but zero recall.	Trusting a high accuracy on a rare-event problem.	A rare class (2%, 5%) and an accuracy near 1 minus that rate.
7	Admissible heuristics in A* search	A* is guaranteed optimal only if its heuristic never overestimates the remaining cost.	Believing a stronger (overestimating) heuristic keeps optimality.	A heuristic that "sometimes overestimates."
8	Time-based splits for forecasting	For time-ordered data, train on the past and test on later periods; random shuffling leaks the future into training.	Random 80/20 splits on time series.	Daily or monthly data and a forecasting goal.

Course scope (public description): Artificial Intelligence Optimization for Computer Scientists guides students through the implementation and optimization of artificial intelligence (AI) solutions for various applications. Through extensive research, students will explore different AI approaches and determine the most applicable solutions for specific scenarios. Practical, hands-on exercises will enable students to implement and rigorously test AI solutions, thus honing their skills in optimizing AI models for enhanced performance and efficiency. Additionally, this course delves into creating data assumptions and interpretations that are crucial for predictive analytics and future data forecasting. Finally, students will adapt and extend AI solutions to address diverse application scenarios, ensuring their readiness to tackle real-world challenges in AI optimization and deployment. Introduction to Artificial Intelligence for Computer Scientists is a prerequisite to this course.